

FlightGear 2024.1.2

F-15EX



User Manual



*Richard Harrison, Enrique Laso (3d), Nikolai V. Chr., Jimmy L. Miles (EX code and 3d interiors),
Robert Andrews (EX interiors modeling)*



TABLE OF CONTENTS

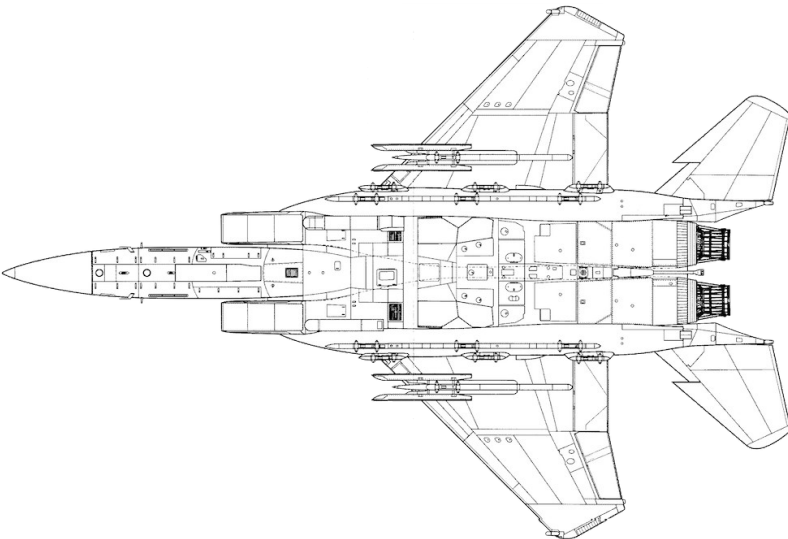
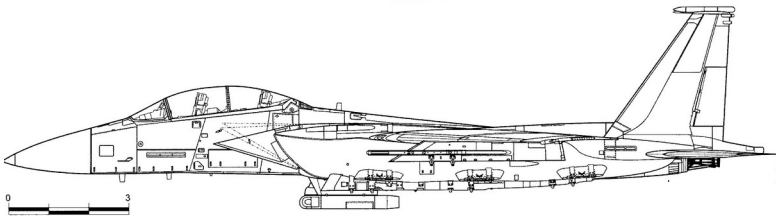
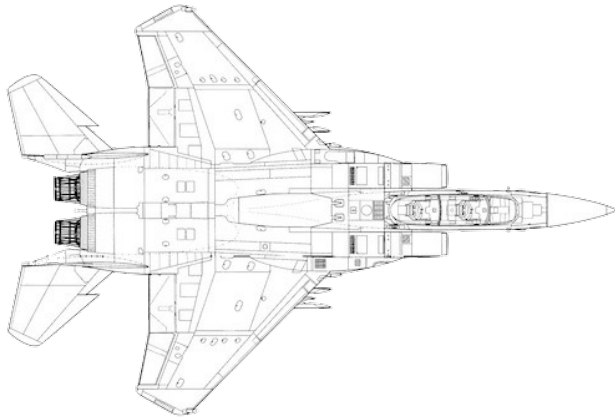
- Chapter 1 : Introduction
 - 1.1 Aircraft Description
 - 1.2 F-15C Background
 - 1.3 F-15E Background
 - 1.4 F-15EX Program
- Chapter 2 : Front Cockpit
 -
- Chapter 3 : Rear Cockpit (WSO)
 -
- Chapter 4 : Normal Procedures
 - 4.1 Introduction
 - 4.2 Aircraft Startup
 - 4.2.1 Before Engine Start
 - 4.2.2 Engines Start
 - 4.2.3 Systems and Sensors
 - 4.2.4 INS/GPS Alignment
 - 4.2.5 Other steps
 - 4.3 Taxiing and Takeoff
 - 4.4 Landing
 - 4.4.1 Descent Check
 - 4.4.2 Landing Using an Overhead Brake
 - 4.4.3 Before Landing Checks
 - 4.4.4 Crosswind Landing
 - 4.4.5 After Landing
 - 4.4.6 Aircraft Shutdown
- Chapter XX : Air to Ground Weapons
 - X.1 Introduction
 - X.2 Air to Ground Weapons
 - X.2.1 Unguided bombs
 - X.2.2 Laser-Guided bombs
 - X.2.3 JDAM's (GPS/INS Bombs)
 - X.2.4 Close-Range Missiles
 - X.2.5 Stand-Off Air-to-Land Missiles
 - X.2.6 Anti-Ship Missiles
 - X.3 Air to Ground Loadouts
 - X.4 Programmable Armament Control Set (PACS)
 - X.5 Max-Performing
 - X.5.1 GBU-39A SDB
 - X.5.2 AGM-154A JSOW
 - X.5.3 AGM-84E SLAM

Chapter 1: Introduction





1.1 Aircraft Description



Crew

2 (a pilot and a Weapon System Officer)

Length - Wingspan - Height

63'9.6" – 42'9.6" – 18'6"

Empty Weight

31,700 lbs (with CFTs)

30,000 lbs (without CFTs)

Max T/O Weight

80,000 lbs

Max Payload

29,000 lbs

Max Speed (High Alt)

1,500 kts

Max Speed (Low Alt)

800 kts

Service Ceiling

65,000'

Rate Of Climb

50,000 fpm

Power Plant

2 x F110-GE-129

Dry Thrust: 17,155 lbs each

AB Thrust: 29,500 lbs each

Thrust to Weight ratio

.85-1.22 (depends on payload)



1.2 F-15C Background

The F-15C has often been labeled as the greatest fighter aircraft in the world. Designed to counter the exaggerated capabilities of the Soviet MiG-25 Foxbat, the F-15C has been the backbone of U.S. air defense for three decades. The F-15C, equipped with improved avionics and weapons over the original F-15A, has scored over 100 air-to-air victories in the service of Israel, Saudi Arabia, and the U.S. without suffering any losses.



The F-15C rules the Beyond Visual Range arena (BVR). No slouch in a dogfight, the F-15C excels at finding targets, positively identifying them as hostile, and engaging them with AIM-120C AMRAAM missiles before the enemy can respond.

The F-15's versatile pulse-Doppler radar system can look up at high-flying targets and down at low-flying targets without being confused by ground clutter. It can detect and track aircraft and small high-speed targets at distances beyond visual range down to close range, and at altitudes down to tree-top level. The radar feeds target information into the central computer for effective weapons delivery. For close-in dogfights, the radar automatically acquires enemy aircraft, and this information is projected on the heads-up display.

The Eagle can also be deadly in the close-in dogfight. The AIM-9M Sidewinder, a reliable weapon that has soldiered on since the 1960's, does not have the high off-boresight capability of recent Russian heat-seeking missiles. F-15C drivers should generally favor the higher-speed "energy fight" in favor of the low-speed turning duel, especially against nimble adversaries. However, in a slow-speed fight, the Eagle's large rudders are a powerful tool in the hands of a skilled pilot.



Chapter 4: Normal Procedures

A clean parked F-15EX with wheel chocks and a fire extinguisher



4.1 Introduction

This fourth chapter will cover normal procedures, focusing mostly on the front cockpit. The rear cockpit is relatively the same and all procedures can be executed from both cockpits. In this chapter, images and indications of switches and knobs will be from both cockpits.

The chapter will cover the following subjects:

- Aircraft startup
- Taxiing and Takeoff
- Landing
- Aircraft Shutdown

Each of the subjects described above has a separate checklist available at the end of this manual, in Appendixes.



4.2 Aircraft Startup

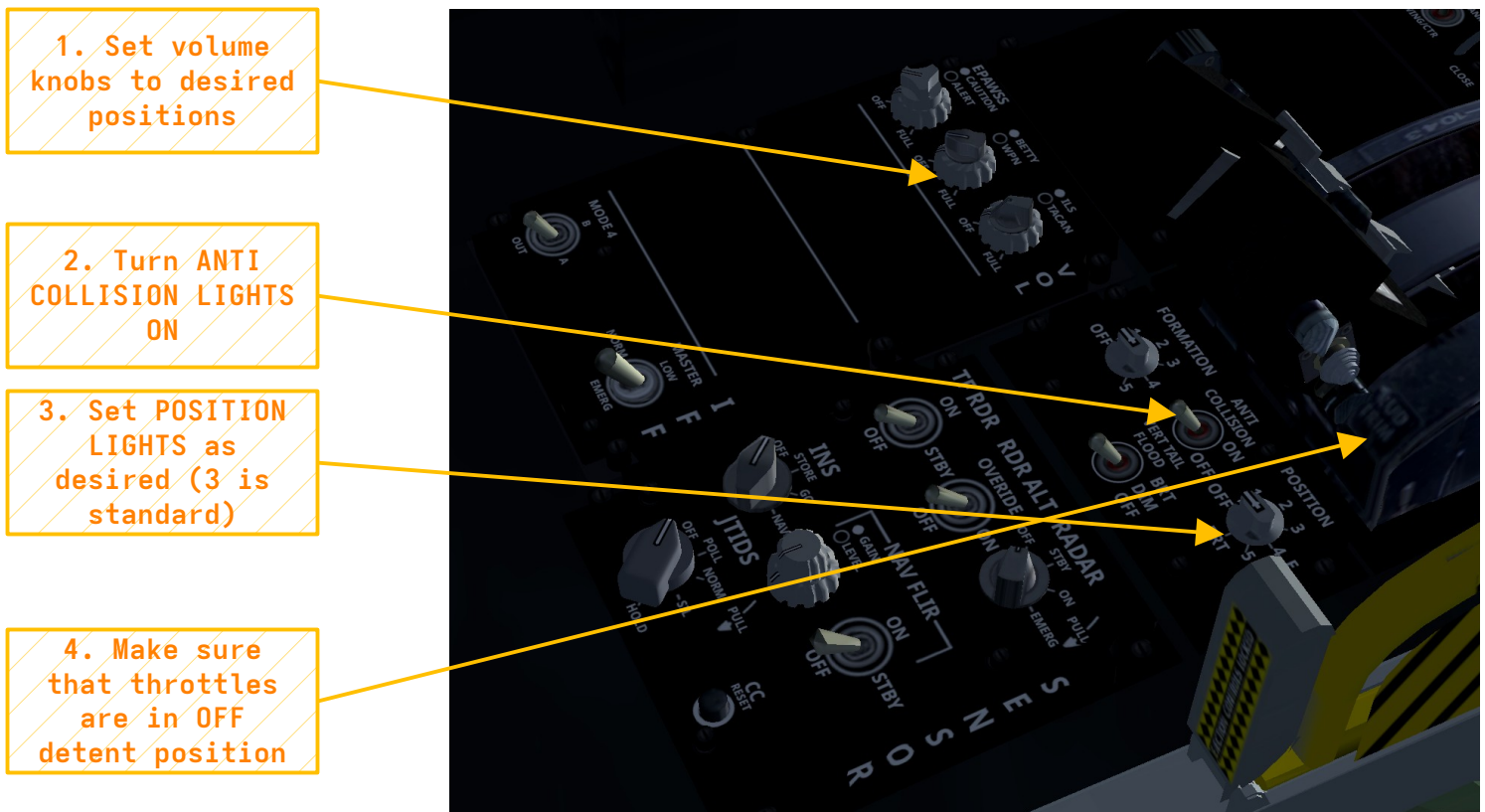
The F-15EX is a relatively easy aircraft to prepare for flight, with most of the old F-15E and F-15C switches being removed and made automatic in the Central Computer. The whole procedure, including full INS alignment, can take between 10 minutes with only necessary actions performed to 20-30 minutes with all BIT and system checks.

You can use the **Quickstart** button on the **Boeing F-15EX Eagle II Config Panel** for the aircraft startup to run instantaneously.

Relevant checklist for this part of the manual can be found in the *Appendixes*.

4.2.1 Before Engine Start

LEFT PANEL #1



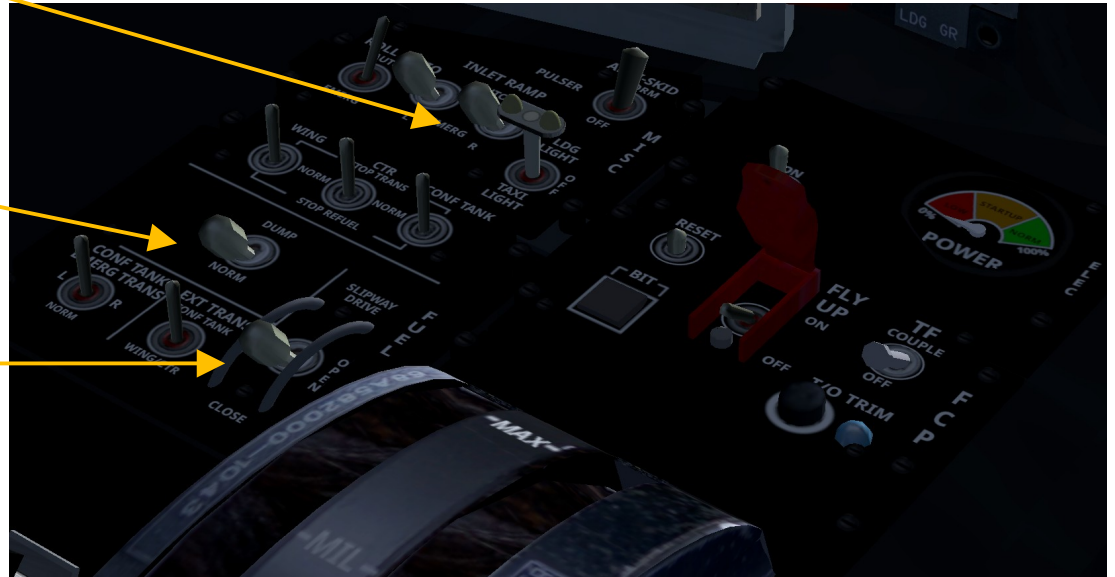


LEFT PANEL #2

6. Set both
INLET RAMP
SWITCHES to AUTO

7. Make sure
FUEL DUMP SWITCH
is to NORM

8. Make sure
REFUEL DOOR is
to CLOSE



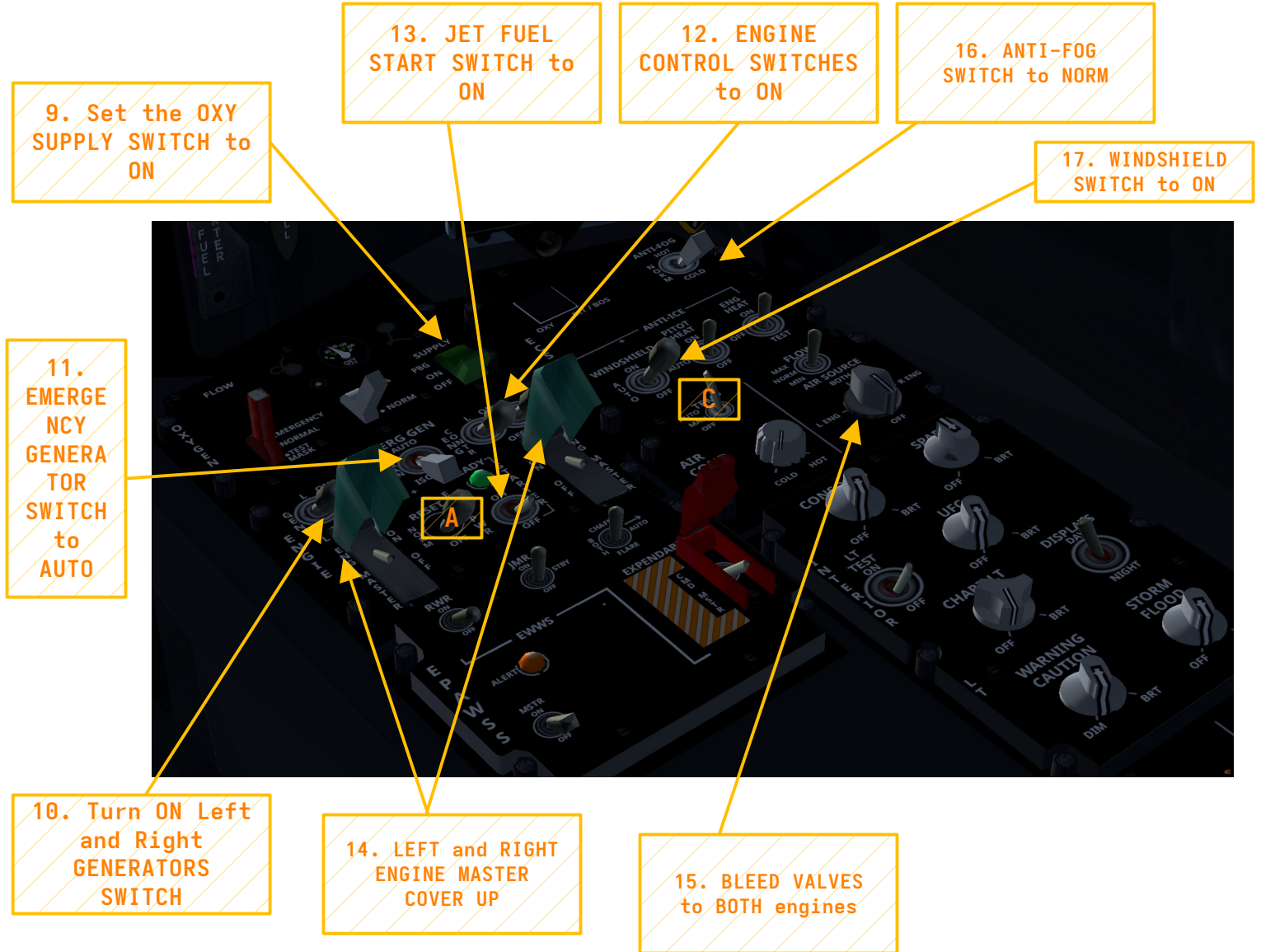
Setting INLET RAMP SWITCHES allows the ramps to drop down once the necessary RPM is reached (around 50%). Below is an example of the right ramp being already down.



A Air-Combat-ready F-15EX with its right engine started and its left engine shutdown

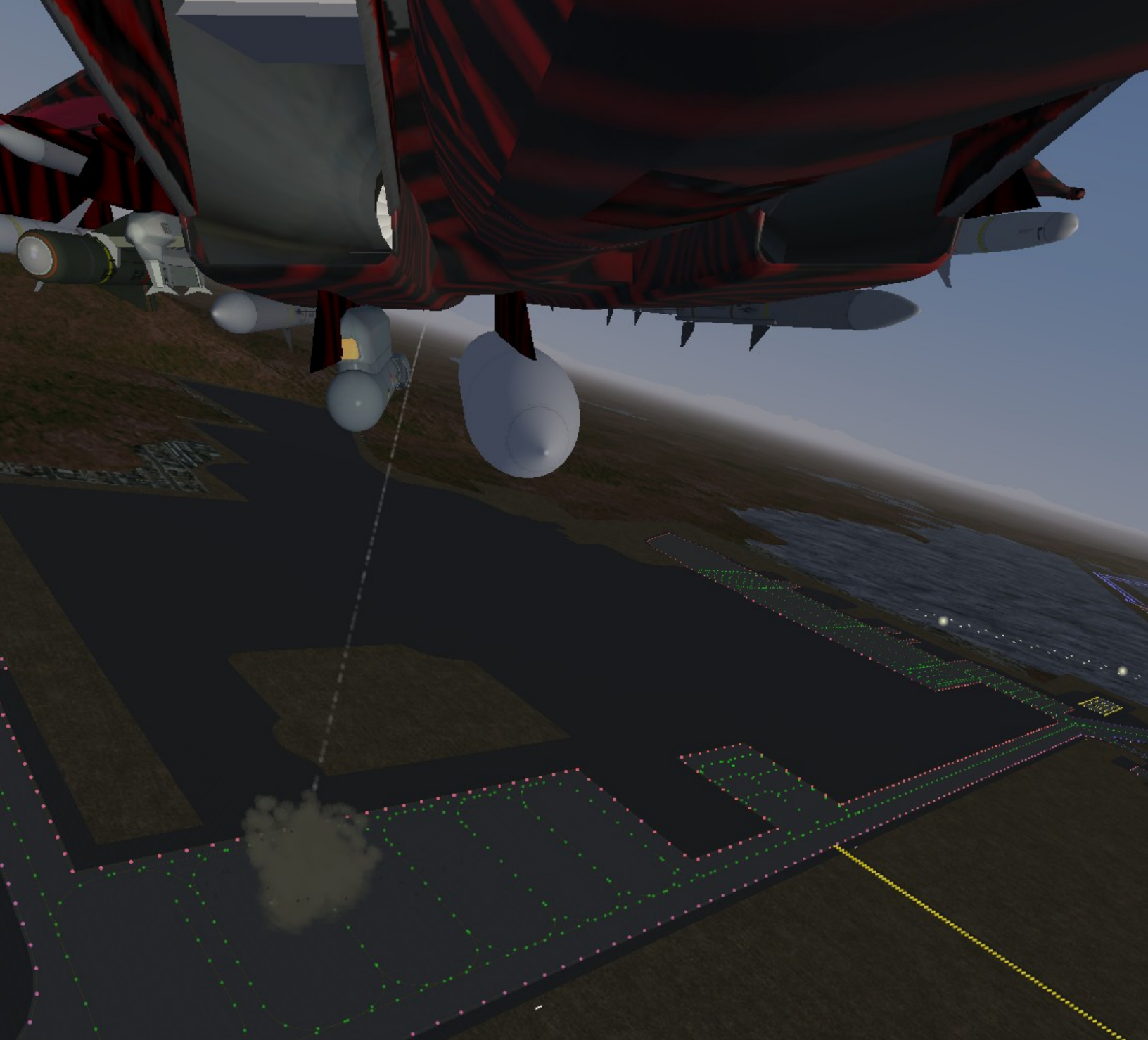


RIGHT PANEL #1



EXTERNAL POWER SWITCH (A) should not be used until external power is physically connected to the aircraft (Ground Power Unit and External Air demand on the Boeing F-15EX Eagle II Config Panel deploys a G.P.U. to get external power and external air (for JFS)). The use of ground power and air allows an engines start without JFS and allows the BIT's (Built-In-Test) to operate as soon as ground power is connected, without waiting for the own aircraft's generators to be ON by one of the two engines getting near 60% RPM.

ECS SWITCH (B) is important, as without setting it up the avionics won't receive proper cooling and may shut down or even get damaged. *This is not modeled in this simulated model.*



Chapter XX: Air to Ground Weapons

A F-15EX in the Nevada Test and Training Range (NTTR) launching AGM-65D Mavericks



X.1 INTRODUCTION

Despite the F-15EX being the replacement for the old 1970's F-15C, with A/A as a primary role, it's also based on the F-15E and is designed to perform multirole missions and excel in that. It's capable of using most of the – and the latest – USAF inventory, as well as SEAD ordnances, of which the F-15E didn't. The F-15EX is a very potent and versatile aircraft.

X.2 Air to Ground Weapons

The F-15EX can carry every type of air to ground weapons that's ever been invented by mankind. From dumb bombs, smart cluster munitions, laser guided bombs, bunker busters (the GBU-28 5,000-pounder), modern JDAM-LSR bombs, and more modern smart weapons such as the JSOW, JASSM, SLAM, but also anti-ship missiles, such as the Harpoon, or the stealthy long-range LRASM. It's also capable of carrying and deploying SEAD and DEAD armaments, such as the TV, IR and Laser Mavericks, and finally the very-capable AGM-88E AARGM Anti-Radiation missile, upgrade of the AGM-88C HARM, enhanced with an active radar for terminal guiding, or MADDOG deployments, a GPS/INS guidance for Pre-Planned targets, providing a high resistance to SAM's countermeasures.

The F-15EX's Mission Computer, world's fastest mission computer ever installed on a fighter jet, is capable of storing pre-planned GPS/JDAM targets in 4 different "Mission Sets", each of them containing 40 "Mission Programs", which can be edited/created in-flight, or loaded through a Data Cartridge. This means that the EX is capable of storing up to 160 GPS/JADM targets. But its different sensors, the Targeting Pod, the powerful Air to Ground Radar and also the Helmet Mounted Cueing Display are all capable of providing precise and quick target points, allowing the deployment of smart weapons in-flight, without the being planned and loaded through the Data Cartridge.

Smart Weapons take the EX to another level, transmitting data while homing and flying toward their target back to the Link 16 channel, allowing live ordnances to be monitored in real time, and precision-adjusted.



Note that the weapons that this F-15EX simulated model is able to carry, haven't been necessarily officially cleared to be used on the actual EX, though they've been seen loaded on ground-parked F-15s, sometimes foreign F-15s.

X.2.1 Unguided Bombs

Also known as free-fall bombs, or dumb bombs, would, in a real-life scenario - even 20 or 30 years ago - never be used. Unguided bombs were only used more than 50 years ago, in earlier F-15E Blocks. Though, the EX can carry and deploy those. For the sake of having more weapons available in this simulated model, and for the fun that dumb bombs can bring in-game, unguided bombs are fully modeled in this model.

In real life, since unguided are dumb bombs (meaning they don't carry a computer on them), aren't detected by the Mission Computer - not like smart weapons as the JDAM's for example - they must be pilot-programmed to the Mission Computer, or through the Data Cartridge. But in the simulated model, there is no such process, dumb bombs are recognized automatically. Arming times are also un-editable in-flight - but Smart Weapons can - but in this simulated model, arming times can be edited in-flight in the PACS

Mark-82 Bomb

A general purpose, low drag, free-fall bomb widely used by NATO forces - in the past.

Type	Low drag	Warhead (size)	192 lbs
Mass	500 lbs	Length	7'3"
Smart Weapon	no	Data telemetry	no
Short name	M82	Full name	Mark-82 General Purpose Bomb
Max G	1.6 G's	Operational Range	8 NM





Mark-82 AIR Bomb

Variant of the Mark-82 bomb with air inflatable retarder, slowing down its descent rate and allowing more time for the delivering aircraft for safe escape. Commonly used on low-pass bombs, especially on runways.

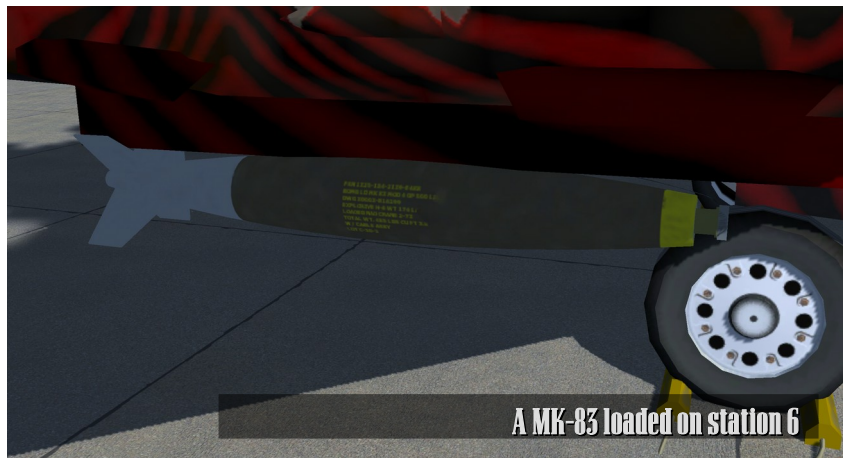
Type	High drag	Warhead (size)	192 lbs
Mass	500 lbs	Length	7'3"
Smart Weapon	no	Data telemetry	no
Short name	M82A	Full name	Mark-82 Air Inflatable Retarder
Max G	1.6 G's	Operational Range	8 NM



Mark-83 Bomb

General purpose, low drag, free fall bomb. In between the Mk-82 and the Mk-84 in terms of size and weight.

Type	Low drag	Warhead (size)	445 lbs
Mass	1014 lbs	Length	9'8"
Smart Weapon	no	Data telemetry	no
Short name	M83	Full name	Mark-83 General Purpose Bomb
Max G	1.8 G's	Operational Range	8 NM

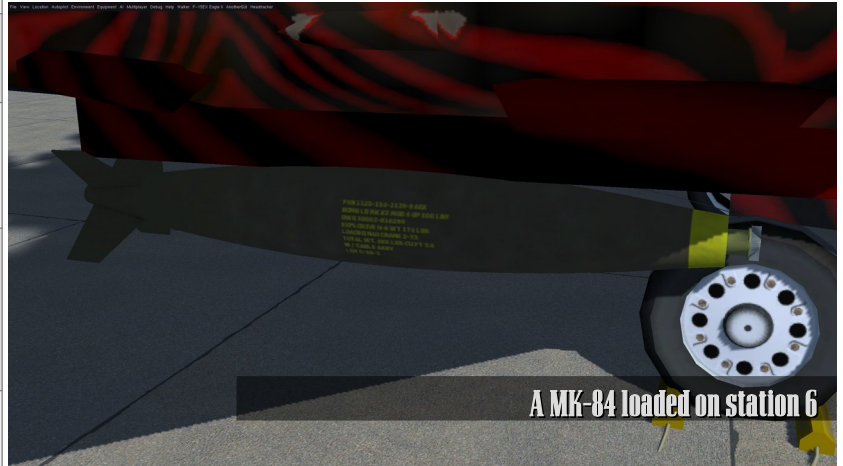




Mark-84 Bomb

General purpose, low drag, free fall bomb. Largest of the Mark 80 series of weapons.

Type	High drag	Warhead (size)	945 lbs
Mass	2,039 lbs	Length	12'7"
Smart Weapon	no	Data telemetry	no
Short name	M84	Full name	Mark-84 General Purpose Bomb
Max G	2.0 G's	Operational Range	8 NM

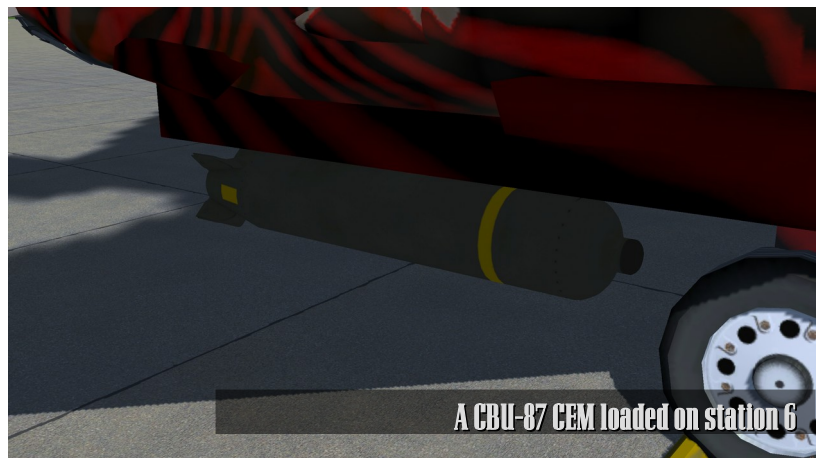


A MK-84 loaded on station 6

CBU-87 Combined Effects Munition

Each CBU-87 holds 202 bomblets. It can be dropped at any altitude and any airspeed. After deployment, it begins to spin. At a preset altitude the canister breaks open and the submunitions release, covering a large area.

Type	Unguided Cluster	Warhead (size)	202 bomblets (128 lbs)
Mass	950 lbs	Length	7'7"
Smart Weapon	no	Data telemetry	no
Short name	CB87	Full name	CBU-87 Combined Effects Munition
Max G	1.8 G's	Operational Range	7 NM



A CBU-87 CEM loaded on station 6



X.2.2 Laser-Guided Bombs

GBU-12 Paveway II

The GBU-12 is a Mark-82 general purpose bomb, with the addition of a nose-mounted laser seeker and fins for guidance.

Type	Laser Guided	Warhead (size)	190 lbs
Mass	610 lbs	Length	10'7"
Smart Weapon	yes	Data telemetry	no
Short name	GB12	Full name	GBU-12 Paveway II
Max G	1.6 G's	Operational Range	15 NM



GBU-24 Paveway III

The GBU-24 is an enhanced GBU-10 - a Mark-84 with a laser guidance kit - with better guiding capabilities and therefore increased range.

Type	Laser Guided	Warhead (size)	945 lbs
Mass	2,257 lbs	Length	14'5"
Smart Weapon	yes	Data telemetry	no
Short name	GB24	Full name	GBU-24 Paveway III
Max G	2.0 G's	Operational Range	15 NM



GBU-28 Bunker Buster

A 5,000-pounder, bunker-busting bomb designed during Operation Desert Storm to enable the allied forces to penetrate hardened Iraqi command centers. A later variant had itself added a GPS/INS guidance kit alongside the laser guidance, similarly to the GBU-38. It's been reported to penetrate 6 yards and 2 feet of reinforced concrete.

Type	Laser Guided and GPS/INS guided	Warhead (size)	675 lbs
Mass	5,000 lbs	Length	19'1"
Smart Weapon	yes	Data telemetry	no
Short name	GB28	Full name	GBU-28 Bunker Buster
Max G	2.5 G's	Operational Range	5 NM



X.2.3 JDAM's (GPS/INS Bombs)

GBU-31 JDAM

A Mark-82 bomb with a BLU-109 penetrator and GPS/INS guidance kit, capable of penetrating concrete shelters and other hardened structures before exploding.

Type	GPS/INS guided	Warhead (size)	945 lbs
Mass	2,039 lbs	Length	12'5"
Smart Weapon	yes	Data telemetry	no
Short name	GB31	Full name	GBU-31 JDAM
Max G	2.0 G's	Operational Range	15 NM



GBU-32 JDAM

A Mark-83 bomb with a BLU-109 penetrator and GPS/INS guidance kit, capable of penetrating concrete shelters and other hardened structures before exploding.

Type	GPS/INS guided	Warhead (size)	445 lbs
Mass	1,015 lbs	Length	9'8"
Smart Weapon	yes	Data telemetry	no
Short name	GB32	Full name	GBU-32 JDAM
Max G	3.5 G's	Operational Range	15 NM

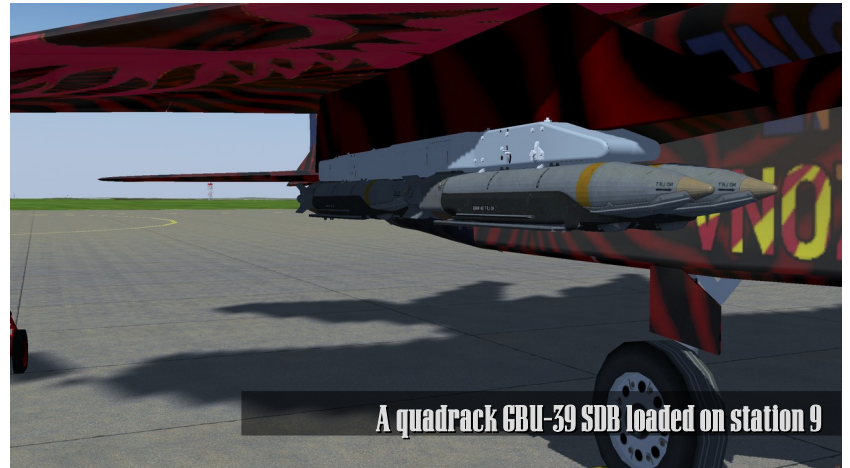




GBU-39 Small Diameter Bomb I

A small-diameter bomb, with deployable wings, small and light enough to be carried by four on a BRU-61 rack, capable of reaching far-off targets, and penetrating concrete.

Type	GPS/INS guided	Warhead (size)	206 lbs
Mass	285 lbs	Length	5'11"
Smart Weapon	yes	Data telemetry	yes
Short name	GB39	Full name	GBU-39 Small Diameter Bomb I
Max G	3.75 G's	Operational Range	70 NM



GBU-54 Laser JDAM

A Mk-82 bomb equipped with a GPS/INS guidance kit, as well as a nose-mounted laser seeker, making it a more versatile weapon capable of hitting moving targets.

Type	Laser-GPS/INS guided	Warhead (size)	190 lbs
Mass	558 lbs	Length	7'8"
Smart Weapon	yes	Data telemetry	no
Short name	GB54	Full name	GBU-54 Laser JDAM
Max G	1.6 G's	Operational Range	15 NM





CBU-105 Sensor Fuzed Weapon

Each CBU-105 holds 10 Infrared-guided bomblets. It can be dropped at any altitude and any airspeed. After deployment, it begins to spin. At a preset altitude the canister breaks open and the submunitions release, covering a large area. The CBU-105 canister is also guided, using a JDAM GPS/INS kit.

Type	GPS/INS-guided Cluster	Warhead (size)	10 IFR bomblets (65 lbs)
Mass	934 lbs	Length	7'8"
Smart Weapon	no	Data telemetry	no
Short name	CB105	Full name	CBU-105 Sensor Fuzed Weaon
Max G	1.8 G's	Operational Range	10 NM



A duorack CBU-105 SFW loaded on station 9

AGM-154 JSOW (Joint Stand-Off Weapon)

The AGM-154A is a long-range JDAM capable of gliding towards the predefined target. The A and C variant differ in warhead: the A variant offers a load of 154 BLU-97/B Combined Effects Bomb (CEB) munitions, while the C variant offers a BROACH multi-staged single warhead. The C variant also offers a IFR seeker for terminal homing. Though, the USAF has never ordered or used the C variant, only the A variant has been. Only the A variant is available in the simulated model.

Type	GPS/INS glide bomb	Warhead (size)	154 IFR bomblets (65 lbs)
Mass	1,065 lbs	Length	13'4"
Smart Weapon	no	Data telemetry	no
Short name	AG154A	Full name	AGM-154A-1 JSOW
Max G	3.0 G's	Operational Range	70 NM



A duorack AGM-154 JSOW loaded on station 9



X.2.4 Close-Range Missiles

AGM-65 B/D Maverick

The AGM-65 Maverick dates back to the 1970's - as of the F-15 - , used humongous amounts of times in every American War since then. Designed for close-range air support, it's an IFR/TV - guided missile, capable of hitting mobile targets and penetrating armored tanks. It can reach speeds of 600 kts. The AGM-65B is TV-guided, while the AGM-65D is Infrared-guided, the D variant being night-and-fog-capable and the B not. The variants be visually be indentified: the B variant is gray while the D is green.

Type	TV/IFR Missile	Warhead (size)	126 lbs
Mass	485 lbs	Length	8'2"
Smart Weapon	yes	Data telemetry	no
Short name	AG65B/D	Full name	AGM-65B /D Maverick
Max G	25 G's	Operational Range	12 NM



LAU-68C / M151 (Hydra-70 Rocket Launcher)

The LAU-68C or M151 is a Hydra-70 launcher, capable of containing 7 Hydras. The Hydras are unguided A/G rockets designed to destruct armored-buildings and tanks.

Type	Unguided rocket launcher	Warhead (size)	X7 Hydra (10 lbs)
Mass	311 lbs	Length	8'2"
Smart Weapon	no	Data telemetry	no
Short name	LAU68/M151	Full name	LAU-68C Rocket Launcher
Max G	10 G's	Operational Range	2 NM





X.2.5 Stand-Off Air-to-Land Missiles

AGM-84E SLAM (Standoff Land Attack Missile)

The AGM-84E SLAM (Standoff Land Attack Missile) is a long-reach A/G cruise missile, GPS/INS-guided, equipped with an AGM-65D's IFR seeker for terminal homing. Its infrared seeker allows for hit precision if the given Latitude/Longitude target isn't precise enough, or if a manual correction must be done by the launching aircraft (a.k.a. man-in-the-loop technology). The IFR terminal homing can also be set to automatic. Its data-link capability allows GPS/INS correction from the launcher aircraft.

Type	GPS/INS Cruise Missile	Warhead (size)	500 lbs
Mass	1,190 lbs	Length	14'10"
Smart Weapon	yes	Data telemetry	yes
Short name	AG84E	Full name	AGM-84E SLAM
Max G	5.0 G's	Operational Range	120 NM





AGM-158A JASSM (Joint Air-to-Surface Stand-off Missile)

The AGM-158A JASSM is a stealthy Air-to-Ground cruise missile, capable of extreme ranges. It's guided by a GPS/INS kit, and it utilizes an IFR seeker for terminal homing and precision, similar to that of the AGM-84E SLAM, unless that the main-in-the-loop capability isn't present on the JASSM. Its data-link capability allows for GPS/INS correction from the launcher aircraft.

Type	GPS/INS Cruise Missile	Warhead (size)	1,000 lbs
Mass	2,150 lbs	Length	14'1"
Smart Weapon	yes	Data telemetry	yes
Short name	AG158A	Full name	AGM-158 A JASSM
Max G	6.0 G's	Operational Range	200 NM





X.2.6 Anti-Ship Missiles

AGM-84D Harpoon

The AGM-84D Harpoon is a long-proven anti-ship missile, considered one of the world's best. It uses a GPS/INS guidance to guide itself to an area where enemy ships are thought to be, and then uses an internal radar to seek for ships and home onto them. It skims as low as 60 feet to be as stealthy as possible – though in modern cases it won't help – and targets the ship's hull's center for a maximum effect.

Type	GPS/INS and Active Radar Anti-Ship Missile	Warhead (size)	500 lbs
Mass	1,190 lbs	Length	14'10"
Smart Weapon	yes	Data telemetry	no
Short name	AG84D	Full name	AGM-84D Harpoon
Max G	5.0 G's	Operational Range	120 NM



An AGM-84D Harpoon loaded on station 9



AGM-158C LRASM (Long-Range Anti-Ship Missile)

Base off of the AGM-158A JASSM, the AGM-158C LRASM is a stealthy long-range anti-ship cruise missile, being the best response to enemy fleets in a modern scenario. As the AGM-84D Harpoon, it uses a GPS/INS guidance to guide itself onto a fleet or ship. But instead of an active radar, it fuzes different sensors: data-link data, its own infrared seeker, and onboard Artificial Intelligence. As the Harpoon, it also does sea-skimming, as low as 20 feet, but, unlike the Harpoon, it can terrain-follow, meaning it won't crash into any terrain if launched in a scenario with islands. Its 1,000-pound warhead makes it extremely deadly to ships' hulls.

Type	GPS/INS and IFR Anti-Ship Cruise Missile	Warhead (size)	1,000 lbs
Mass	2,150 lbs	Length	14'1"
Smart Weapon	yes	Data telemetry	yes
Short name	AG158C	Full name	AGM-158 C LRASM
Max G	6.0 G's	Operational Range	500 NM



An AGM-158C LRASM loaded on station 9



X.3 Air to Ground Loadouts

The F-15EX is capable of carrying a mix of air-to-air and air-to-ground in many different configurations. It can also carry up to 3 700-gallon fuel tanks (a.k.a. Bags) on its wings' and centerline's pylons. It can also carry an AA/AAQ-13 LANTIRN Navigation Pod, permitting night missions, on its right intake's pylons, and an AA/AAQ-33 Sniper XR Laser Pod, allowing the designation of ground targets at day or night or foggy day, capturing information for T00 (Target Of Opportunity) ammunitions, capable of lasing up to 75 NM away. It can also carry theIRST (Infrared Search and Track) Legion Pod, allowing the stealthy-tracking of targets in a clear sky, permitting the EX to launch AIM-120D's at targets, without the target getting a missile launch warning or a radar emitter RWR warning. The CFT's (Conformal Fuel Tanks), dating back from the F-15E, greatly broaders the amount of weapon the F-15EX is capable of carrying, adding 6 new pylons for light bombs or AMRAAM's, and allowing it to carry heavier ordnances on the conformal-fuselage stations. Though, whether an F-15EX should carry CFT's or not during a mission is an important decision that shouldn't be overthought. On the first hand, CFT's greatly broader the amount of damage the F-15EX can deal to its enemies, increasing its range with the nearly 7,000 lbs of fuel capacity added. Whereas on the other hand, the CFT's create significant drag and weight which cannot be ignored in a close-range A/A combat or a dogfight, which cannot be "supressed in flight" due to the CFT's uncapability to be jettisonned, unlike the external 700-gallon Bags. Depending on the requirements of the mission, you should decide whether to install CFT's or not.



X.4 Programmable Armament Control Set (PACS)

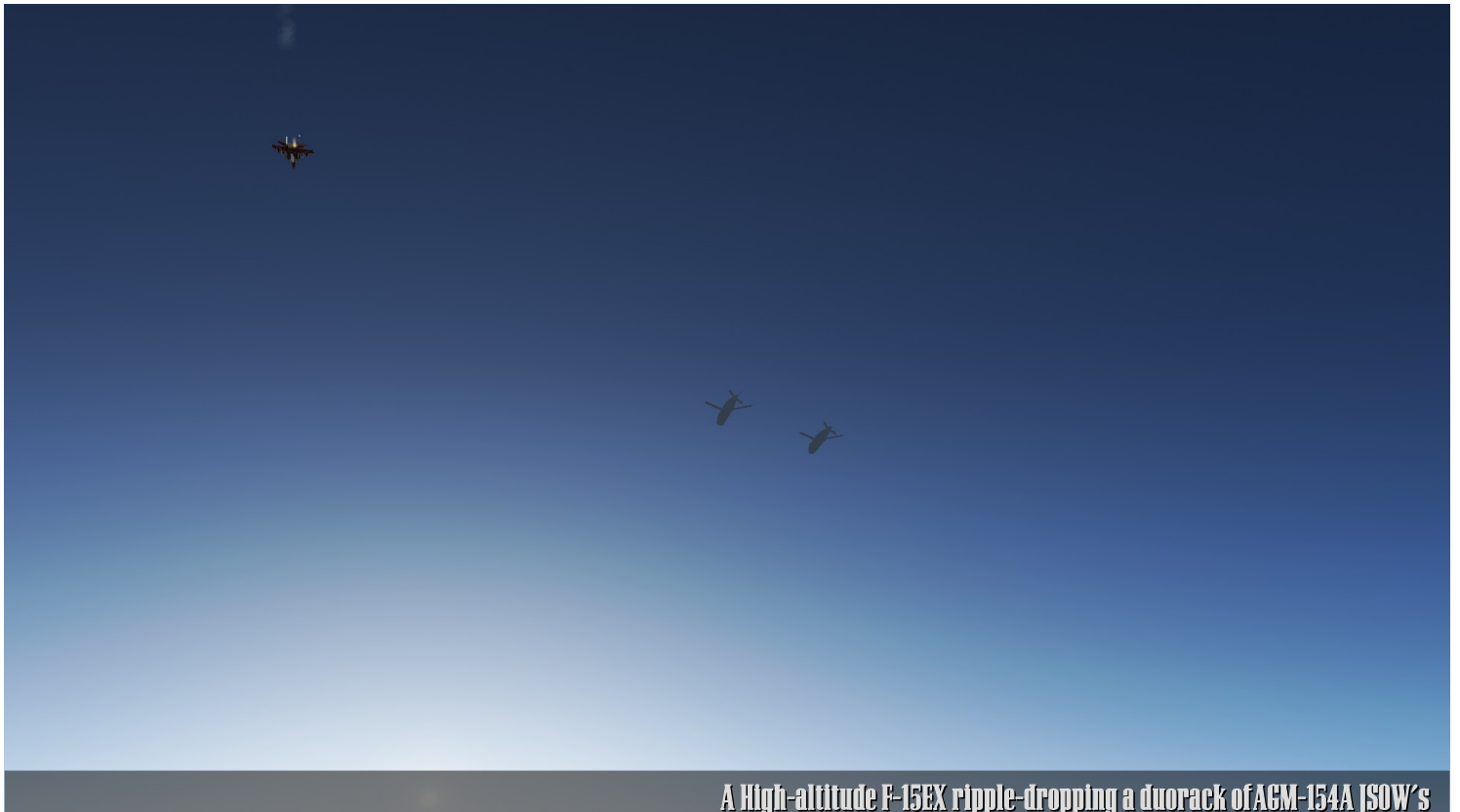
The PACS provides weapons monitoring, as well as display / management capabilities. It's used for selection, pre-launch preparation, launch and jettison of both air-to-air and air-to-ground ordnances. Both aspects will be covered seperately, in their own sub-chapter.



X.5 Max-Performing

Knowing which launch parameters are best for your Air-to-Ground ordnances is key to success. Each ordnance requires different launching speeds, altitudes, and precision (in the case of GPS/INS guided ammunitions) depending on the target's distance, in order to perform that ordnance at its maximum and to have full expected effects on target. Some ordnances require less strict launch parameters, being powered by their own rocket or jet engines, but for bombs, even the slightest offset in launch parameters can significantly alter the result. In this sub-chapter, each ordnance will be given information upon their optimal launch parameters, at each of their different deployment modes. An ordnance's Operational Range, is the maximum range that that ordnance's can theoretically reach in optimal conditions.

This sub-chapter won't mention unguided munitions.



A High-altitude F-15EX ripple-dropping a duorack of ACM-154A JSOW's



X.5.1 GBU-39A SDB

The GBU-39 being a bomb, does not propell itself, but it can perform long-reaching flights due to its small and aerodynamic body, along with deployable wings, generating lift. Its Operational Range is of 70 NM. Though, performing its 70 NM of operational range requires a high altitude and high speed. To perform a 70 NM hit, you'll need to fly at 50,000' MSL, at Mach 2.5. With these parameters, you should get a 100% hit precision, with a Hit Speed of around 531 kts. You can find a table under this piece of text, giving you the performance of the GBU-39A Small Diameter Bomb under different launch parameters. Note that the SDB does not avoid terrain on its own: if the target is in a steep valley, make sure to release from a high altitude (Flight Level 30 to 65).

As the GBU-39A is a GPS/INS guided ordnance with no IFR or TV seeker for terminal homing, the precision of the target's coordinates is critical. 2 decimals of seconds of Latitude and Longitude are required for a precise hit.

	Altitude (% F15 ceiling)	Speed (Mach)	Distance (% opr rng)	Hit Speed (kts)
Envelope 1	FL500 MSL (80%)	2.5	70 NM (100%)	531



X.5.2 AGM-154A JSOW

The AGM-154 being a bomb, does not propell itself, but it can perform long-reaching flights due to its small and aerodynamic body, along with deployable wings, generating lift. Its Operational Range is of 70 NM. Though, performing its 70 NM of operational range requires a high altitude and high speed. To perform a 70 NM hit, you'll need to fly at 50,000' MSL, at Mach 2.5. With these parameters, you should get a 100% hit precision, with a Hit Speed of around 531 kts. You can find a table under this piece of text, giving you the performance of the AGM-154A JSOW under different launch parameters. Note that the JSOW does not avoid terrain on its own: if the target is in a steep valley, make sure to release from a high altitude (Flight Level 30 to 65).

As the AGM-154A is a GPS/INS guided ordnance with no IFR or TV seeker for terminal homing, the precision of the target's coordinates is critical. 2 decimals of seconds of Latitude and Longitude are required for a precise hit.

	Altitude (% F15 ceiling)	Speed (Mach)	Distance (% opr rng)	Hit Speed (kts)
Envelope 1	FL500 MSL (80%)	2.5	70 NM (100%)	531



X.5.3 AGM-84E SLAM

The AGM-84E SLAM has got loose restrictions on its launch parameters, it being powered by its own engine, and being derived from the Harpoon which needed to be launched at 5,000 ft MSL 50 NM from its target. A high speed of at least Mach 1 and an altitude of at least 10 to 15 thousands of feet should be enough to reach a 60 to 80 NM target. Note that the SLAM does not avoid terrain on its own: if the target is in a steep valley, make sure to release from a high altitude (Flight Level 30 to 65). Though, it does maintain level flight at the altitude upon which it was released, meaning that if you release at an altitude higher than the highest ground point of the area (highest mountain or example), and that the target isn't in a steep valley, the SLAM shouldn't hit any piece of land.

As the AGM-84E SLAM is a GPS/INS guided ordnance with an IFR seeker for terminal homing, the precision of the target's coordinates isn't as much critical as pure GPS/INS munitions. Whole decimals of seconds of Latitude and Longitude are enough in order to perform a precise hit (if terminal IFR is ON). 2 decimals of seconds of Latitude and Longitude are required for a precise hit (if terminal IFR is OFF).

	Altitude (% F15 ceiling)	Speed (Mach)	Distance (% opr rng)	Hit Speed (kts)
Envelope 1	FL400 MSL (60%)	1.1	120 NM (100%)	1,620
Envelope 2	FL320 MSL (50%)	1.26	105 NM (85%)	1,399
Envelope 3	FL250 MSL (40%)	1.3	105 NM (85%)	1,141
Envelope 4	FL150 MSL (25%)	1.425	80 NM (65%)	1,095
Envelope 5	FL100 MSL (15%)	1.5	80 NM (65%)	875